

§6.AK Check 10 — api-app Channel Extension Patch (exact, copy-pasteable)

Date: 2026-06-26 **Status:** Ready to apply (logic proven hermetically — see §4 below) **Closes:** the api-app portion of §6.AK-debt (the owed follow-up named verbatim in the live Check 10 comment at `.github/hooks/pre-push:204-206` and in `docs/superpowers/specs/2026-06-26-ak-gate-wiring-integration.md §2.2`). **Companion test:** `tests/cycle-coverage/test_wiring_apiapp.sh` (5/5 PASS against the real gate; mutation-rehearsal proven — §4). **Spec parent:** `docs/superpowers/specs/2026-06-26-ak-cycle-coverage-spec.md §5.4` + `docs/superpowers/specs/2026-06-26-ak-gate-wiring-integration.md §2`.

Classification: project-specific (the insertion is Lava-file-specific; the gate-coverage-must-intersect-cycle-claims pattern is universal per §6.AK).

IMPORTANT — this document does NOT edit the live gate file. It is the precise patch the main stream applies later. The target file (`.github/hooks/pre-push`) is pushed-through-pre-push concurrently by the main stream; editing it here would corrupt an in-flight gate. Apply this patch only when the main stream is quiescent.

0. Why this patch

§6.AK gate-wiring landed in commit 28a8b79b: `.github/hooks/pre-push Check 10` + `scripts/firebase-distribute.sh Phase-1 Gate 7` both enforce `scripts/check-cycle-coverage.sh`.

The asymmetry the live code documents (`.github/hooks/pre-push:204-206`):

```
# CLIENT-channel scope (mirrors Check 7); api-app advances are gated at
# distribute time by firebase-distribute.sh Gate 7 (owed
```

§6.AK-debt follow-up

to extend Check 10 to the api-app channel).

- **CLIENT** channel (.lava-ci-evidence/distribute-changelog/firebase-app-distribution): a last-version-{debug,release} advance is gated at **both** push time (Check 10) AND distribute time (Gate 7).
- **api-app** channel (.lava-ci-evidence/distribute-changelog/firebase-app-distribution-api-app): a last-version-{debug,release} advance is gated **only at distribute time** (Gate 7 reads \$CHANGELOG_DIR, which app-resolves to the api-app dir for --app api-app). At **push time** the api-app pointer advance is invisible to the hook.

The gap: a commit that advances the api-app last-version-debug/-release pointer with failing cycle-coverage is *accepted by the push hook* and is only caught later at distribute time. That is exactly the 1076 incident shape (627a0d58: a pointer-advance "ready to distribute" signal landing without covering device evidence) — pushed first, blocked later, instead of refused at the earliest gate. This patch closes the api-app push-time gap so BOTH channels are gated identically at push time.

1. The patch — wrap Check 10's per-channel body in an outer channel loop

1.1 Strategy

The cleanest, lowest-risk approach: the existing Check 10 already hardcodes a single `ak_chan_dir` and then runs the full for `ptr` in `last-version-debug last-version-release; do ... done advance-detect-and-gate` loop against it. Wrap that exact body in an **outer for `ak_chan_dir` in <client> <api-app>; do ... done** loop. Zero logic inside the body changes — every `git show "$sha^:$ak_ptr_path"`, snapshot lookup, `--evidence-dir="$ak_chan_dir"`, and `--head="$sha"` invocation already keys off `$ak_chan_dir`, so the body is already channel-agnostic. This is the minimal diff that proves identical behavior on both channels.

1.2 Exact current block (anchor — .githooks/pre-push lines 195-239)

The block as committed (28a8b79b), reproduced for unambiguous anchoring:

```

# ===== Check 10: $6.AK cycle-coverage on distribute-
# advancing commits =====
# Closes the pre-push portion of $6.AK-debt (companion to
# firebase-
# distribute.sh's Phase-1 Gate 7). When a commit ACTUALLY
# advances

# last-version-debug or last-version-release (NEW > OLD -
# same detection as

# Check 7), a pointer-advance signifies "ready to
# distribute"; the $6.AK
# cycle-coverage gate MUST then pass for that version: every
# CHANGELOG-claimed
# user-visible fix needs an executed+PASSED covering device
# Challenge in the
# $6.Z evidence file. Pushing the advance without covering
# device evidence is
# the 1076 incident shape (627a0d58: C00-only gate shipped
# broken flows).

# CLIENT-channel scope (mirrors Check 7); api-app advances
# are gated at
# distribute time by firebase-distribute.sh Gate 7 (owed
# $6.AK-debt follow-up
# to extend Check 10 to the api-app channel).
ak_chan_dir=".lava-ci-evidence/distribute-changelog/firebase-
app-distribution"
for ptr in last-version-debug last-version-release; do
    ak_ptr_path="${ak_chan_dir}/${ptr}"
    git diff-tree --no-commit-id --name-only -r "$sha" 2>/dev/
    null | grep -qF "$ak_ptr_path" || continue
    ak_old=$(git show "$sha^:$ak_ptr_path" 2>/dev/null | tr -d
    ' \n' || echo "0")
    ak_new=$(git show "$sha:$ak_ptr_path" 2>/dev/null | tr -d '
    \n' || echo "0")
    [[ "$ak_new" =~ ^[0-9]+$ && "$ak_old" =~ ^[0-9]+$ &&
    "$ak_new" -gt "$ak_old" ]] || continue
    case "$ptr" in
        last-version-debug)    ak_channel="debug" ;;
        last-version-release)  ak_channel="release" ;;
        *)                     ak_channel="debug" ;;
    esac
    ak_vname=""
    ak_snapshot=$(ls "${ak_chan_dir}"/*-${ak_new}.md 2>/dev/
    null | grep -v '\-test-evidence\.md$' | head -1 || true)
    if [[ -n "$ak_snapshot" ]]; then
        ak_vname=$(basename "$ak_snapshot" -${ak_new}.md)
    else
        ak_vname=$(git show "$sha:app/build.gradle.kts" 2>/dev/
        null | grep -oE 'versionName[[:space:]]*=[[:space:]]*"([^\
        +"' | head -1 | sed 's/\..*"([^\"]*)\.*/\1/' || echo "")
    fi
    [[ -n "$ak_vname" ]] || continue

```

```

if [[ -x ./scripts/check-cycle-coverage.sh ]]; then
    ak_rc=0
    ./scripts/check-cycle-coverage.sh \
        --version="${ak_vname}-${ak_new}" \
        --channel="$ak_channel" \
        --evidence-dir="$ak_chan_dir" \
        --head="$sha" \
        --strict >/dev/null 2>&1 || ak_rc=$?
    if [[ "$ak_rc" -ne 0 ]]; then
        violations+=("$sha: §6.AK violation – advances $ptr $
        {ak_old}→${ak_new} (${ak_vname}) but the cycle-coverage
        gate exits $ak_rc. A CHANGELOG claim lacks a covering
        executed+PASSED device Challenge (or the §6.Z evidence /
        cycle-coverage-map is missing / stale / wrong-SHA). Run
        the missing device Challenge(s) on the gate, OR strike the
        unverified claim(s) from CHANGELOG.md (§6.AK clause 6),
        before pushing the pointer advance.")
    fi
fi
done

```

1.3 The replacement block (paste in place of the block in §1.2)

Replace the entire block above with this. **Three deltas only**, all marked <<< api-app extension:

1. the comment's CLIENT-channel scope ... owed §6.AK-debt follow-up note becomes a BOTH-channel scope note;
2. the single ak_chan_dir=...firebase-app-distribution assignment becomes an outer for ak_chan_dir in ...firebase-app-distribution ...firebase-app-distribution-api-app; do;
3. a matching done closes the outer loop after the existing inner for ptr ... done.

The 31 lines of inner body (for ptr in ... done) are **byte-identical** to §1.2 — only re-indented by two spaces to sit inside the outer loop.

```

# ===== Check 10: §6.AK cycle-coverage on distribute-
# advancing commits =====
# Closes the pre-push portion of §6.AK-debt (companion to
# firebase-
# distribute.sh's Phase-1 Gate 7). When a commit ACTUALLY
# advances

# last-version-debug or last-version-release (NEW > OLD –
# same detection as

# Check 7), a pointer-advance signifies "ready to
# distribute"; the §6.AK

```

```

# cycle-coverage gate MUST then pass for that version: every
# CHANGELOG-claimed
# user-visible fix needs an executed+PASSED covering device
# Challenge in the
# $6.Z evidence file. Pushing the advance without covering
# device evidence is
# the 1076 incident shape (627a0d58: C00-only gate shipped
# broken flows).
# BOTH-channel scope: client (firebase-app-distribution) AND
# api-app # <<< api-app extension
# (firebase-app-distribution-api-app). The api-app channel is
# also gated at # <<< api-app extension
# distribute time by firebase-distribute.sh Gate 7; gating it
# here too closes # <<< api-app extension
# the push-time gap ($6.AK-debt api-app follow-up). Each
# channel's pointer + # <<< api-app extension

# snapshot + $6.Z evidence live under its own dir, so the
# same body keys off # <<< api-app extension
# $ak_chan_dir
# unchanged.
# <<< api-app extension
for ak_chan_dir in \
".lava-ci-evidence/distribute-changelog/firebase-app-
distribution" \
".lava-ci-evidence/distribute-changelog/firebase-app-
distribution-api-app"; do # <<< api-app extension
for ptr in last-version-debug last-version-release; do
ak_ptr_path="${ak_chan_dir}/${ptr}"
git diff-tree --no-commit-id --name-only -r "$sha" 2>/dev/
null | grep -qF "$ak_ptr_path" || continue
ak_old=$(git show "$sha^:$ak_ptr_path" 2>/dev/null | tr
-d ' \n' || echo "0")
ak_new=$(git show "$sha:$ak_ptr_path" 2>/dev/null | tr -d
' \n' || echo "0")
[[ "$ak_new" =~ ^[0-9]+$ && "$ak_old" =~ ^[0-9]+$ &&
"$ak_new" -gt "$ak_old" ]] || continue
case "$ptr" in
last-version-debug) ak_channel="debug" ;;
last-version-release) ak_channel="release" ;;
*) ak_channel="debug" ;;
esac
ak_vname=""
ak_snapshot=$(ls "${ak_chan_dir}"/*-${ak_new}.md 2>/dev/
null | grep -v '\-test-evidence\.md$' | head -1 || true)
if [[ -n "$ak_snapshot" ]]; then
ak_vname=$(basename "$ak_snapshot" -${ak_new}.md)
else
ak_vname=$(git show "$sha:app/build.gradle.kts" 2>/dev/
null | grep -oE 'versionName[[:space:]]*=[[:space:]]*"([^\
+]" | head -1 | sed 's/.*"\([^"]*\)".*/\1/' || echo "")
fi
[[ -n "$ak_vname" ]] || continue
if [[ -x ./scripts/check-cycle-coverage.sh ]]; then
ak_rc=0

```

```

./scripts/check-cycle-coverage.sh \
--version="${ak_vname}-${ak_new}" \
--channel="$ak_channel" \
--evidence-dir="$ak_chan_dir" \
--head="$sha" \
--strict >/dev/null 2>&1 || ak_rc=$?
if [[ "$ak_rc" -ne 0 ]]; then
    violations+=("$sha: §6.AK violation – advances $ptr $
{ak_old}→${ak_new} (${ak_vname}) on $
{ak_chan_dir###*/} but the cycle-coverage gate exits
$ak_rc. A CHANGELOG claim lacks a covering executed+PASSED
device Challenge (or the §6.Z evidence / cycle-coverage-
map is missing / stale / wrong-SHA). Run the missing
device Challenge(s) on the gate, OR strike the unverified
claim(s) from CHANGELOG.md (§6.AK clause 6), before
pushing the pointer advance.")
fi
done
done
<<< api-app extension (closes the outer channel loop)

```

The only body-line change beyond re-indentation is the `violations+=` message: on `{ak_chan_dir###*/}` is appended so the operator sees WHICH channel (client vs api-app) tripped. This is a message-only delta; the exit/reject behavior is identical.

1.4 Apply mechanics (unambiguous)

- The block sits inside the per-SHA for sha in \$range; do ... done loop (lines 58–261 of the current file) and appends to the violations array exactly like every other Check. The outer channel loop adds one nesting level; the existing two-space body indentation becomes four-space — re-indent the 31 inner-body lines accordingly (already done in §1.3).
 - Net structural change vs. §1.2: + for ak_chan_dir in ... do before the for ptr loop, + done after it, the comment note swap, and the `###*/` message suffix. Nothing else.
 - Independent verification of correctness: `bash -n .githooks/pre-push` (syntax) after applying, then `bash tests/cycle-coverage/test_wiring_apiapp.sh` (must stay 5/5) and `bash tests/cycle-coverage/test_wiring.sh` (must stay 5/5 — the client path is unchanged).
-

2. §6.N.1.2 + Check 9 obligations on the GRAFT COMMIT (do not skip)

The commit that applies this patch will itself trip the existing hook:

- **Check 4 / §6.N.1.2** (.githooks/pre-push:92-126) fires because the diff touches .githooks/pre-push (in Check 4's gate-shaping list at :97-98). The graft commit body **MUST** carry a Bluff-Audit: stamp that NAMES at least one file in the diff. Cite tests/cycle-coverage/test_wiring_apiapp.sh as the rehearsal. Use the stamp in §3 below.
 - **Check 9 / CM-SCRIPT-DOCS-SYNC** (.githooks/pre-push:241-258) does NOT apply: .githooks/pre-push is not under scripts/*.sh, so no companion docs/scripts/*.md is required for this graft. (If the same commit also touches a scripts/*.sh with an existing docs/scripts/<name>.sh.md, that doc must be updated — but this patch touches only .githooks/pre-push and adds a test under tests/.)
 - **Check 2** (.githooks/pre-push:67-74) fires because the diff adds tests/cycle-coverage/test_wiring_apiapp.sh (matches Test\kt|_test\.go? — NO: the filename is test_wiring_apiapp.sh, which does NOT match (Test\kt|_test\.go)\$, so Check 2 does NOT fire on it). The Bluff-Audit: stamp is still REQUIRED by Check 4 because the diff touches .githooks/pre-push.
-

3. The §6.N.1.2 Bluff-Audit stamp the graft commit **MUST** carry

Bluff-Audit: .githooks/pre-push (§6.AK Check 10 api-app channel extension)

Mutation: swallow the api-app refusal — change the inner
`violations+=("...§6.AK violation...")` push into a
no-op (`:`) so a
failing api-app cycle-coverage gate is accepted at
push time.

Observed-Failure: tests/cycle-coverage/test_wiring_apiapp.sh reports

"FAIL [apiapp_advance_debug_fail] exit=0 (expected 1)" and
"FAIL [apiapp_advance_release_fail] exit=0 (expected 1)" — the
wrapper even

prints "pre-push Check 10 (api-app) OK — no §6.AK violation"
AFTER a failing

gate (the exact 1076 push-time-gap shape the extension
closes);

"cases passed: 3 cases failed: 2 / RESULT: FAIL".

Reverted: yes

This stamp is verbatim what the companion test's own Bluff-Audit: header documents and what the captured mutation rehearsal produced (§4).

4. Proof the extension logic is sound (hermetic, no real file edited)

tests/cycle-coverage/test_wiring_apiapp.sh generates a STUB pre-push wrapper whose grafted block mirrors §1.3's api-app body (the same Check 10 logic keyed off the api-app ak_chan_dir + per-ptr channel), invokes the REAL scripts/check-cycle-coverage.sh against synthetic api-app §6.Z evidence + cycle-coverage-map fixtures (HEAD + now injected via LAVA_CYCLE_COVERAGE_HEAD / LAVA_CYCLE_COVERAGE_NOW_EPOCH), and asserts both pointers (debug + release), both verdicts (covered + uncovered), plus the no-advance bypass:

```
test: §6.AK CHECK-10 api-app channel EXTENSION (pre-push push-time
gate)
  PASS [apiapp_advance_debug_pass] exit=0 (expected 0) + matched
'Check 10 (api-app) OK'
  PASS [apiapp_advance_release_pass] exit=0 (expected 0) +
matched 'Check 10 (api-app) OK'
  PASS [apiapp_advance_debug_fail] exit=1 (expected 1) + matched
'§6.AK violation'
  PASS [apiapp_advance_release_fail] exit=1 (expected 1) +
matched '§6.AK violation'
  PASS [apiapp_no_advance] exit=0 (expected 0) + matched 'Check
10 (api-app) OK'
-----
cases passed: 5    cases failed: 0
RESULT: PASS – pre-push Check 10 api-app channel extension proven
against the real gate
```

Mutation rehearsal (swallow the api-app refusal → negative cases flip): with the inner violations+= changed to a no-op, the two negative cases flip to FAIL and the wrapper prints "Check 10 (api-app) OK" after a failing gate:

```
  PASS [apiapp_advance_debug_pass] exit=0 (expected 0) + matched
'Check 10 (api-app) OK'
  PASS [apiapp_advance_release_pass] exit=0 (expected 0) +
matched 'Check 10 (api-app) OK'
  FAIL [apiapp_advance_debug_fail] exit=0 (expected 1) + needle
'§6.AK violation'
    ---- wrapper output ----
    pre-push Check 10 (api-app) OK – no §6.AK violation
  FAIL [apiapp_advance_release_fail] exit=0 (expected 1) + needle
'§6.AK violation'
```



```
----- wrapper output -----
pre-push Check 10 (api-app) OK — no $6.AK violation
PASS [apiapp_no_advance] exit=0 (expected 0) + matched 'Check
10 (api-app) OK'
-----
cases passed: 3    cases failed: 2
RESULT: FAIL — the api-app Check 10 extension logic is unsound; do
NOT graft as-is
```

The extension is therefore falsifiable and sound: it refuses an api-app pointer advance whose cycle-coverage gate fails, and a swallowed refusal is mechanically detectable.

5. What this does NOT change

- scripts/check-cycle-coverage.sh — untouched (already committed + tested; the gate is channel-agnostic).
- scripts/firebase-distribute.sh Gate 7 — untouched; it already gates the api-app channel at distribute time via \$CHANGELOG_DIR.
- Checks 1–9 + the client path of Check 10 in pre-push — untouched; this patch only widens Check 10's channel scan from one dir to two.
- tests/cycle-coverage/test_wiring.sh (the client-channel wiring test) — untouched; it remains the client-path proof, and must stay 5/5 after the graft.